

The cut-off mirrored double cone

Stage 1 as an hourglass — and a cone the model shapes itself

E-080 • design proposal • 15 September 2026 • **nothing here is measured yet**

The deck turns Chris's four points into a specification: a waist instead of a tip, a mirrored cone of targets, a random mix of four tasks, and a cone whose shape the model learns per region — one direction per flow, one scale per channel.



One cone, five seeds, one clear reading

E-069 • the cone-native codec

- 30 days of dots around a North Atlantic cell, one 3×3 patch at the present, targets at +1 and +2 pentads
- History did **not** put the cell's own velocity into the embedding — five seeds, two objectives, refuted
- History **did** buy persistent fields (hidden sea-surface height at ~0.36 of the bar vs 0.75–0.90) and a 7–8 % better one- and two-pentad forecast

E-071 • cone v2

- Global, harmonic climatology, profile tokens — the geometry's known faults, fixed
- Reach from the fastest mechanism × 1.5 — a hemisphere by lag 3, so dots go log-radial

The open question v2 left: a cone wide enough to never exclude a driver samples the near field at one dot per thousands of kilometres.

Past cone → waist → future cone



- **Past cone** (lags $-6 \dots -1$): read as input, dots at the per-lag ellipse plus the anchor column
- **Waist** (lag 0): a disc of several cells — the cut-off tip — read as input and reconstructed; the embedding z summarises the waist
- **Future cone** (leads $+1 \dots +6$): never read; sampled the same way and used as targets only
- **Point-mirrored**: the future cone at lead $+\ell$ is the past cone at lag $-\ell$ reflected through the anchor

A gradient does not exist within a single cell

- The waist is a disc of radius r_0 around the anchor, sampled as a small sunflower: ~13 cells at the default (one centre + 12), never fewer than the 3×3 patch's nine
- Read as input at lag 0 (values + observed flags) and reconstructed as the anchor family's target
- **What the waist buys:** thermal wind is a density gradient; geostrophy is a sea-surface-height slope — both need neighbours at the same instant, not in the past
- The waist has the same shape parameters as the cone at lag 0 (orientation, two semi-axes), so a boundary current can have an elongated waist along the flow
- The embedding z summarises the waist; one z per anchor, as today

≈ 13

cells at the waist

one centre + 12, never fewer than the nine of the 3×3 patch

1 z

per anchor, as today

so stage 2 and every archived probe see the same kind of object

Sampled like the past cone, read by nobody

+1 ... +6 pentads

30 days ahead, matching the 30 days of history

targets only

nothing in the future cone is ever read by the encoder

frozen at evaluation

the fixed v2 hourglass scores every arm and every seed

- Leads +1...+6 pentads (30 days ahead), the mirror image of the past cone through the anchor: centre $c(+\ell) = -c(-\ell)$, same axes and orientation at the same $|\ell|$
- Dots are **targets only**: each is a decoder query (channel, Δx , Δy , lead, depth) $\rightarrow (\mu, \log \sigma^2)$, scored with the Gaussian negative log-likelihood already in `cone_codec.py`
- The anchor column at every lead is always among the targets — the fixed-frame forecast stage 2 needs
- Per-lead and per-family loss, each reported against its own predict-the-mean bar (the E-069b rule)
- Evaluation geometry is **frozen** (the fixed v2 hourglass), whatever the training cone learns — so every number stays comparable across arms and seeds

One model, four questions, drawn at random per example

task	input	targets	share	what it teaches
T1 forecast	waist + past cone	future cone	0.35	the full problem
T2 snapshot forecast	waist only (past cone fully hidden)	future cone	0.15	the built-in twin: what history buys, inside one model
T3 nowcast from history	past cone under a dropout pattern; waist hidden	waist	0.25	the present from the past — persistence, advection, memory
T4 fill-in	waist + past cone, both under a dropout pattern	the hidden waist cells and dots	0.25	E-069b's reconstruction family, kept

Dropout patterns for T3 and T4, drawn independently (each fires with its own probability)

channel drop

at the waist — lag 0 only,
chan_drop_scope lag0

recent-lag band

hide every dot at $\text{lag} \leq \ell_o, \ell_o \in \{1, 2, 3\}$

bearing wedge

hide a 90° sector

random dots

each dot hidden with $p = 0.3$

whole waist

T3 only

Three rules, all learned the hard way in E-069

1 • Knowable targets only

A target channel must be visible somewhere in the input (any lag, any dot). Under the first E-069 plan 80 % of hidden-dot targets broke this and 64 % of the loss was predict-the-mean.

2 • No copy targets

The anchor family scores hidden channels only (anchor_hidden_only); a target visible in the input is a copy, not a question.

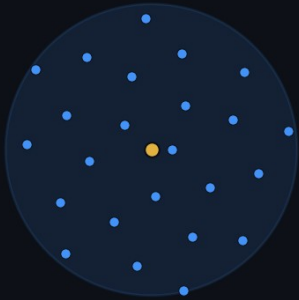
3 • Every family against its own bar

Per (task, family, lead) loss, each divided by its predict-the-mean score; a task is “learned” only where the ratio is below one at held-out anchors.

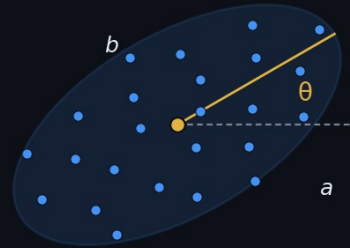
- The task is not told to the model by a token; the masks are visible (the mask_tok / miss_tok distinction stays), and the lead is in every query's coordinates — that is enough
- Loss weights: equal across tasks per example; the shares in the table set the mix, not the weights

Eight numbers per region shape the whole hourglass — and none of them is in kilometres

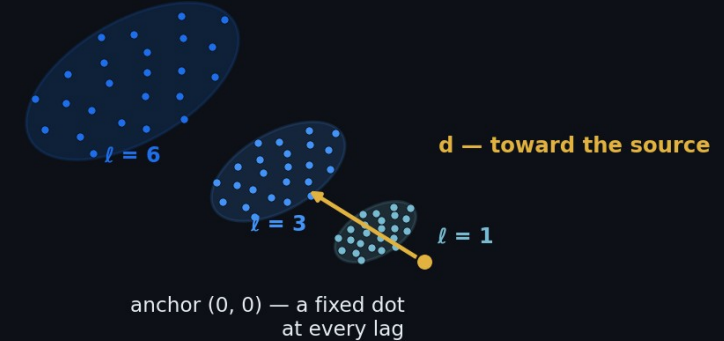
(a) canonical sunflower, unit disc
24 points, golden angle 137.508°



(b) rotated by $\theta = 30^\circ$, stretched to 2:1
semi-axes a, b



(c) the same pattern at three lags: shifted by $\ell \cdot d$, grown with ℓ



$$S = \log \Sigma = m \cdot I + k \cdot [\cos 2\theta, \sin 2\theta; \sin 2\theta, -\cos 2\theta]$$

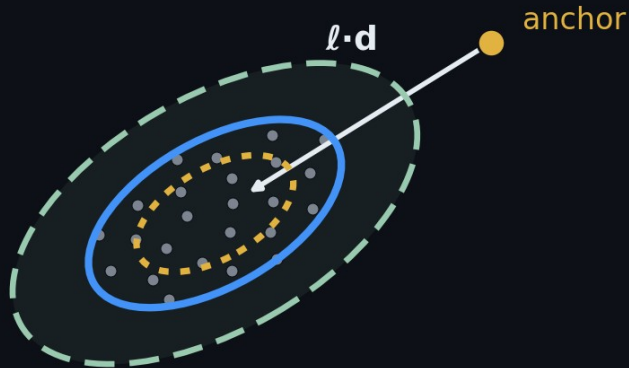
For each region and channel group, the past cone at lag ℓ is a Gaussian footprint centred at anchor + $\ell \cdot d$ with covariance $\Sigma(\ell)$, and the dots are the canonical sunflower mapped through it:

$$\Sigma(\ell) = \Sigma_0 + \ell^2 \Sigma_v \quad p(\ell, i) = \text{anchor} + \ell \cdot d + \Sigma(\ell)^{1/2} \cdot u_i$$

- u_i : the canonical sunflower on the unit disc (Vogel spiral, golden angle — the E-026 pattern in [ml/temporal.py::spiral_offsets](#)), fixed, 24 points, the same at every lag
- d (2 numbers): the drift per pentad — where the footprint's centre moves for each pentad back in time; **the arrow toward the source**. Stored dimensionless: $d = \tanh(x) \cdot v_{\text{design}} \cdot \Delta t$
- $\Sigma_0(3)$: the waist ellipse, stored as its **matrix logarithm** $S_0 = \log \Sigma_0$ — a symmetric 2x2 whose entries are in log-km. Its trace is the size; its traceless part is $(\log \text{aspect ratio}) \times (\cos 2\theta, \sin 2\theta)$. No angle, no 180° wrap; a circle is $S = m \cdot I$
- $\Sigma_v(3)$: the velocity-spread ellipse, also as $\log \Sigma_v$ — how fast the footprint widens with lag (standard deviation $\propto \ell$, as the reach $\propto (1 + \ell)$ in [ml/cone.py::reach_km](#))
- **Why log form**: under Adam every parameter moves about one learning rate per step in its own units. A radius in km moves a fixed number of km (frozen at 850 km, twitchy at 30 km); a log-radius moves a fixed **fraction** — the same behaviour at any size
- **Future cone**: reflected — centre $-\ell \cdot d$, the same $\Sigma(\ell)$; the anchor column (0, 0) is always added. The per-channel scale is the next slide

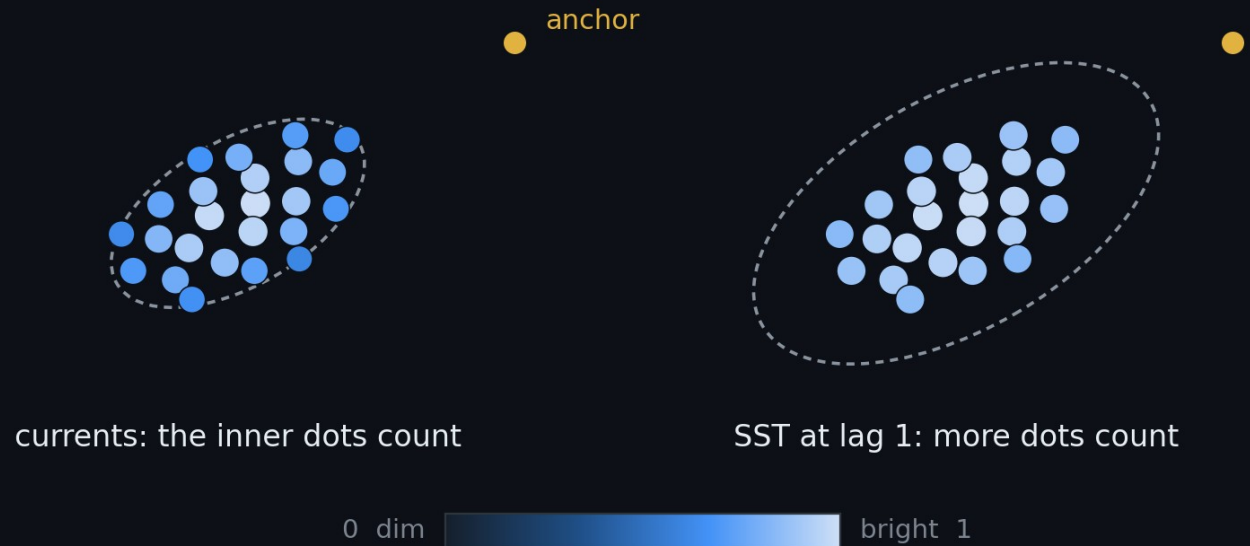
The shape and the drift belong to the flow; only the size belongs to the channel

(a) one dot set at lag 3, three apertures



- currents, sea-surface height · $s = 0$
- - sea-surface temperature at lags 0-1 · $s > 0$ (stirred by the atmosphere)
- - - a channel with $s < 0$

(b) the same dots, coloured by $w(c, i) = \exp(-\frac{1}{2} p^T \Sigma_c(\ell)^{-1} p)$



- **1 • Shared** per region, lag and channel group: the shape $\Sigma(\ell)$ and the drift d of slide 8. Everything the water carries is carried the same way — the arrow toward the source and the elongation along the jet are properties of the **flow**, not of the tracer. *One direction per flow.*
- **2 • Per channel:** one log-scale offset s_c (two for the L-shaped channels: lags 0-1 and 2-6). $S_c(\ell) = S(\ell) + s_c \cdot \ell$, i.e. $\Sigma_c(\ell) = \exp(s_c) \cdot \Sigma(\ell)$ — the same ellipse inflated or shrunk, **never turned**. *One scale per channel.*
- **3 • Initialised from the reach table** in [ml/cone.py::FAMILIES](#): wind stress 500 km at lags 0-1 and nothing beyond; currents and sea-surface height 129.6 km · (1 + ℓ); sea-surface temperature and mixed-layer depth max(that, 500 km) at lags ≤ 1 , then the ocean's growth; land 400 km flat. At step 0 every channel reads the reach the current geometry gives it

One token per dot carries every channel; each channel's entries are weighted by its own aperture before the projection

(c) token anatomy

one per-location token at dot i



- A **per-location token** (E-071 §6.4) carries all channels of a group at one dot position p_i : a vector of values and a vector of observed flags, one token per dot — not one token per channel per dot
- Each channel gets its own weight at that dot, $w_{c,i} = \exp(-\frac{1}{2} \cdot p_i^T \Sigma_c(\ell)^{-1} p_i)$, and it multiplies that channel's value and observed-flag entries **before the projection**. A dot outside channel c 's aperture contributes ≈ 0 for c and still carries the others at full weight
- Worked example**, a dot 400 km from the anchor at lag 1, apertures at their initial values: currents $\sigma = 259 \text{ km} \rightarrow w = \exp(-\frac{1}{2} \cdot (400/259)^2) \approx 0.30$; sea-surface temperature $\sigma = 500 \text{ km} \rightarrow w \approx 0.73$. The same token feeds temperature at 73 % and the currents at 30 %
- The gradient reaches S , d and s_c through w from every dot at once** — not through four neighbouring cells — which is what lets the geometry learn (next slide)
- The fixed safety ring (slide 13) is **not gated**: read at full weight, always. A safety net the model can switch off is not one

A soft aperture over fixed candidate dots in phase 1; the dot table re-baked from the learned ellipses in phase 2

Phase 1 — gate

The candidate dots per region and group are **fixed**: cone v2's log-radial set out to the design reach ([E-071 §4.4](#), 36 per lag) plus the anchor column. The data loader's reads are regular, cacheable, and never depend on the model's weights. Each candidate token is weighted by the per-channel aperture of slide 10, and the ellipses, drift and offsets learn through those weights — from every candidate at once

Phase 2 — harden

Every N steps (2 k to start) the dense dot table per region is re-baked from the current ellipses — the warped sunflower of slide 8, 24 dots per lag — and the loader swaps tables at the epoch boundary. The fixed ring of 24 stays as the safety net; the apertures stay on (they **are** the per-channel scale). Reads remain regular within an epoch

- **Why not read at continuous positions** (the deformable gather of revision 1): it couples the loader's read pattern to the live weights — irregular reads on the sampler's memory-mapped tensor and on the JAX/TPU port of the cone trainer — and its gradient sees only the four cells around each dot. The gate has neither problem, and the per-tensor pyramid of revision 1 is no longer needed
- **Learning rate**: the geometry parameters get their own multiplier ($\times 10$ the codec's to start — the deformable-attention practice, [Zhu et al. 2021](#)), recorded in the run's metrics
- Evaluation geometry stays **frozen** (slide 5) whatever phase training is in

A 1° map per channel group, interpolated to the anchor

- A **1° grid** over the globe ($180 \times 360 = 64,800$ cells); the anchor's parameters are the bilinear blend of the four surrounding map cells — smooth by construction, no seams
- **1° because** the currents that carry the overturning are ~100 km wide (the Florida Current sits in an 80 km strait); a 10° cell would average the Gulf Stream with its recirculation and learn a drift near zero
- **Hierarchical prior:** each map is a coarse 10° map plus a 1° residual, the residual under L2 shrinkage and a Laplacian smoothness penalty — a data-poor cell (ice edge, coast) inherits its neighbourhood's arrow instead of fitting noise. Each 1° cell still sees 16 anchors $\times$ 3,142 pentads $\approx$ 50 k anchor-pentads for eight numbers
- **No Argo ellipse.** The Argo channels are the k nearest profile tokens ([E-071 §3](#), [E-076](#)) — placed where floats were, each carrying its own offset, age and depth. There is no elliptic shape to learn; the search radius stays as specified there. Depth-dependent sourcing (at the RAPID line the upper limb flows north, the Deep Western Boundary Current south) is left to attention over those tokens, which see depth and offset together
- **Optional, phase 2:** a first-harmonic seasonal term on the drift, $d(\tau) = d_0 + d_1 \cos \tau + d_2 \sin \tau$ — the Somali Current reverses with the monsoon

one map per channel group

ocean surface — currents, sea-surface height, sea-surface temperature, mixed layer

atmosphere — drift only at lags 0–1

land — no drift: $v = 0$, axes and angle only

$\approx 1.6 \text{ M}$

geometry parameters, nominal

64,800 cells $\times$ 3 groups $\times$ 8 numbers, about half of it live (ocean maps over ocean, land maps over land), plus ~50 global per-channel scale offsets (slide 9) — cheap, and no compute

Bounded by physics, tied to the fixed cone, started isotropic

Speed cap

$|d| \leq v_{\text{design}} \cdot \Delta t = 5.4 \text{ m/s} \times 5 d = 2,333 \text{ km per pentad (E-071's fastest mechanism} \times 1.5)$, via a tanh parameterisation — the learned cone lives inside the v2 envelope

Log-space, with soft floors

Σ_o , Σ_v and s_c are stored as logarithms (slide 8), so Adam's step is a fixed fraction at any size; the waist's eigenvalues are floored at one cell (28 km) and Σ_v 's at the design speed, by a softplus offset in log space rather than a clip

Prior

Coarse $10^\circ + 1^\circ$ residual, L2 shrinkage and Laplacian smoothness on the residual (slide 12) — one weight, recorded in the run's metrics

Mirror tie, by default

The future cone shares the eight numbers (reflected); an untied arm is the ablation, not the design

Stop-gradient on the target side

The loss reaches the geometry only through the **input** reads; the positions of targets are treated as constants

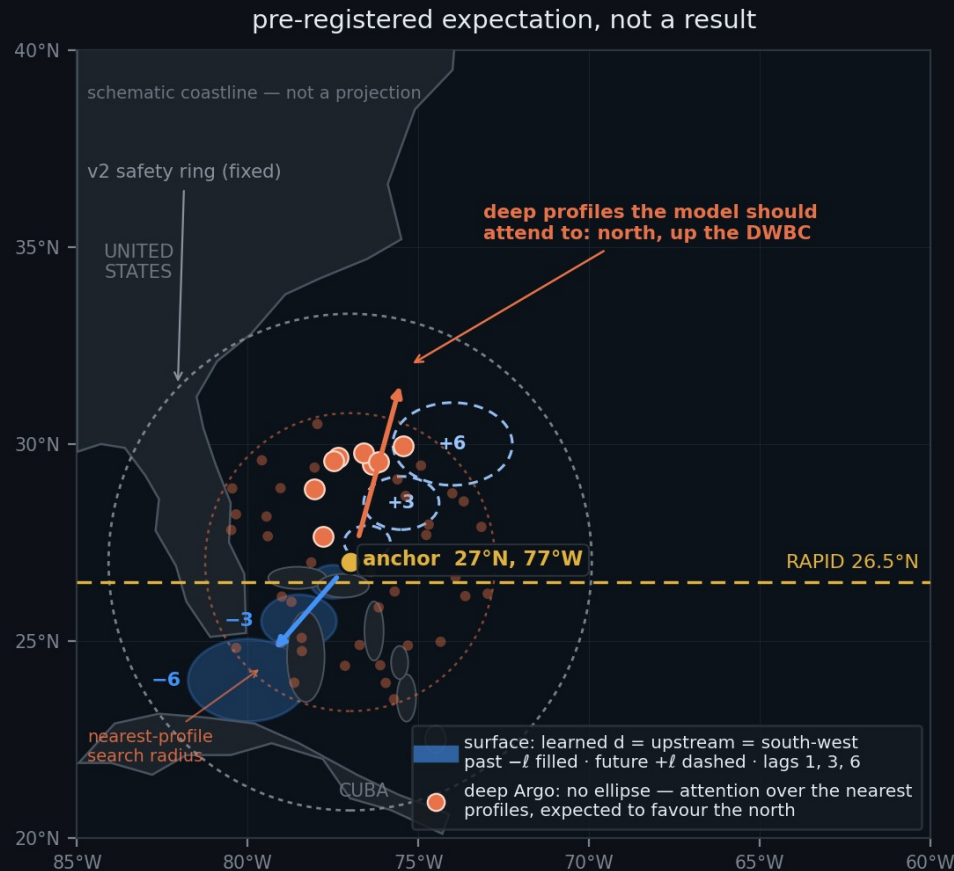
The v2 safety net stays, ungated

In addition to the learned ellipse, each lag keeps one fixed log-radial ring of 24 dots to the antipode, read at full weight — the cone can be directed without asserting that a driver could not have come from elsewhere

Init

$d = 0$; S_o = the log of a 3-cell circle with the measured aspect 0.71 ([ml/measure_flow_anisotropy.py](#)); Σ_v from the 0.3 m/s of [ml/cone.py::FAMILIES](#); s_c from the same table per channel — the fixed hourglass exactly, so step 0 of the learned arm is the control

One anchor on the western boundary: a learned surface cone, and deep profiles read from the other side



- **Surface group:** the learned d should point south-west (up the Florida Current, toward the Loop Current); the future cone leans north-east along the Gulf Stream
- **Deep Argo:** no ellipse to learn. The profile tokens within the search radius carry their own offset and depth; the model should put its attention on the deep (> 1000 dbar) profiles **north** of the anchor — up the Deep Western Boundary Current — and on the shallow ones to the south
- Neither is programmed: these are the **pre-registered expectations** — a geometry check for the surface, an attention check for the deep
- If the learned surface drift correlates with $-u_{\text{clim}}$ (the climatological surface current, reversed) over the world ocean, the cone learned “upstream” from the data alone

A learnable cone can learn to ask easy questions

The loophole: if target positions could move, the gradient would move them to smooth water — a lower loss that means nothing.

Fix 1 • stop-gradient on target positions

Slide 13. The geometry is trained only by “which inputs help”, never by “which targets are easy”.

Fix 2 • the mirror tie

The future cone cannot be tuned separately from the past cone.

Fix 3 • frozen evaluation geometry

Held-out numbers are always scored on the fixed hourglass, for every arm.

Fix 4 • the aperture never touches the loss weights

In phase 1 the targets are the fixed mirrored candidate set with fixed weights; a model that shrinks its aperture cannot thereby shrink the set of questions it is scored on.

Residual second-order effect: as the input cone moves, the mirrored targets move with it. Monitored, not assumed away — report the target-variance under the learned cone vs the fixed cone at every eval; a drop > 10 % is a flag.

Tokens, gathers, parameters — the same regime as v2

	E-069 (measured)	this design (budget)
waist	42 patch tokens (14 ch × 3×3)	13 per-location tokens carrying all surface channels
past cone	706 dot tokens	phase 1: 6 lags × (1 + 36 candidates) × 4 groups ≈ 888 · phase 2: 6 × (1 + 24 ellipse + 24 ring) × 4 ≈ 1,176
Argo	192 (32 × 6, column only)	26 profile tokens (E-071 §3)
targets	42 + 84	256 dot queries per example, drawn from the future cone and the hidden waist / dots (as n_dot_queries today)
geometry parameters	0	≈ 1.6 M nominal, ~half live (slide 12); no compute
extra per-tensor artefact	—	none (revision 1's pyramid is gone); the per-region dot tables re-baked at epoch boundaries are small

- Perceiver cost is linear in tokens: ~1.2× (phase 1) to ~1.6× (phase 2) E-069's encoder time, inside v2's own 2.1× estimate; the aperture weights are one elementwise multiply per token
- Training cost per seed: E-069 seeds ran under one 4090-hour on the North Atlantic; budget 2–3 4090-hours per seed on the global tensor

Three arms, three seeds, four read-outs, one verdict rule

A0 • fixed

The hourglass with the v2 geometry (isotropic ellipses, ring, mirror), no learning — the control, and the eval geometry for everyone

A1 • learned

The eight numbers per region plus the per-channel scales, gate-then-harden (slide 11), initialised at A0

A2 • learned, primed

As A1, but **d** initialised at $-u_{\text{clim}} \cdot \Delta t$ (the climatological surface current, reversed) for the ocean surface group — tests whether the gradient can find upstream on its own (A1) or only keep it (A2)

Held fixed: same tensor (family 7.2, global), same 7 M codec, 20 k steps, frozen protocol (train ≤ 2020 ; 2008–09, 2016–17, 2021–24 held out), 3 seeds each.

R1 T1 future-cone loss per lead vs bar • **R2** the T1 – T2 gap (what history buys) • **R3** drift-vs-upstream correlation, world ocean, $|u_{\text{clim}}| > 0.2$ m/s

R4 the western-boundary depth check — surface drift south-west (geometry); attention mass on deep (> 1000 dbar) profile tokens north of the anchor, on shallow ones south (attention diagnostic)

R5 target-variance flag (slide 15)

Verdict: A1 adopted if it beats A0 on R1 at every lead ≤ 3 , paired at all three seeds, with R5 clean. $A2 > A1$ with $A1 \approx A0$ means the gradient cannot find upstream — then geometry is set from climatology, not learned. Neither $\rightarrow$ keep A0 and move to a dynamic (per-example) cone. **First ablation after the arms:** A1 without the far ring — does the ring pay for its tokens?

Five steps, each with its test

in ml/cone.py · ml/cone_sampler.py · ml/cone_codec.py · ml/train_cone.py

- 1 **Waist + future cone in the geometry (cone.py)** waist_dots, future_dots = reflected inner_dots; coverage_report extended with the mirror identity (future set = -past set) — pure numpy, CPU test
- 2 **The task mix (cone_codec.py::default_plan → plan_v3)** T1-T4 with shares and the five dropout patterns; test: every target's channel is visible somewhere in the input, for 10 k drawn examples, zero violations
- 3 **The aperture gate (cone_codec.py)** per-channel weights $w_{\{c,i\}}$ on the per-location candidate tokens, ring ungated; test: with every $s_c \rightarrow +\infty$ the forward pass equals the ungated model bit-for-bit, and the finite-difference gradient of the loss w.r.t. S_0, Σ_v, d, s_c matches autograd
- 4 **The parameter maps and the harden step (cone_codec.py::ConeGeometry, cone_sampler.py)** 1° maps per group as coarse 10° + residual, matrix-log storage, the log-space floors, the prior, mirror, stop-gradient, init = A0; the sampler's dot table becomes a per-region lookup swapped at epoch boundaries; test: the table re-baked at init equals step 1's fixed dots to < 0.1 cell, and a swap mid-run changes no read of the current epoch
- 5 **Read-outs in the trainer (train_cone.py)** per-task bars, R3 drift maps exported as data/cone_drift_<group>.json for the app's Cones tab, R4 attention-by-depth-and-bearing at the western-boundary anchors, R5 flag; then the three arms as recipes f7l2-hourglass-{fixed,learned,primed}

Each step lands on main with its test before the next starts; step 4 is the only one that touches the data path, and only at epoch boundaries.

Two settled, six still open — all reversible

Settled 15 Sep (Chris): 1° parameter maps with the coarse-plus-residual prior; **no Argo ellipses** — the nearest-profile tokens stay as E-071 §3 specifies them.

Future horizon

+6 pentads to mirror the past; or shorter (+1...+2 as E-069) with denser targets?

Task shares

0.35 / 0.15 / 0.25 / 0.25 — any of them a lever you want set differently?

Prior strength on the 1° residual

One weight, to be set by held-out loss on the training years — or fixed by hand for the first arm?

The v2 safety ring

Kept at every lag (24 tokens per lag per group); dropping it halves the past-cone tokens but restores v2's “too narrow” risk

Seasonal drift term

Deferred to phase 2; or in from the start (monsoon regions)?

Harden interval and geometry LR

N = 2 k steps and the geometry learning-rate multiplier **×10** — first guesses, both recorded per run

Sources

the record this deck stands on

- **E-069 • Two stencils, one cone** — [ml/plans/E069_cone_codec.md](#); results in [ml/EXPERIMENTS.md#e-069](#) (five seeds, H1 refuted; what history buys)
- **E-069b masking-plan fix** — [ml/cone_codec.py::default_plan](#) (chan_drop_scope, anchor_hidden_only), [ml/train_cone.py](#)
- **E-071 • Cone v2** — [ml/plans/E071_cone_v2.md](#) (§3 profile tokens, §4 design speeds and log-radial dots, §4.5 the split in time, §6.4 per-location tokens)
- **E-076 • Family 8 nearest observations** — [ml/plans/E076_family8_nearest_observations.md](#) (the (value, Δx , Δy , Δt , n_R) token)
- **The geometry as built** — [ml/cone.py](#) (FAMILIES, reach_km, slots, inner_dots, outer_spiral, coverage_report)
- **E-026 • the sunflower stencil** — [ml/temporal.py::spiral_offsets](#), the pattern the warped sunflower reuses

and on

- **The geometry learning-rate multiplier and soft sampling weights:** [Zhu et al. 2021, Deformable DETR](#)
- **The step-size argument for log-space parameters:** [Kingma & Ba 2015, Adam](#) — the update is $\approx lr$ per step in the parameter's own units
- **The matrix logarithm of a symmetric 2×2 covariance:** $S = m \cdot I + k \cdot [\cos 2\theta, \sin 2\theta; \sin 2\theta, -\cos 2\theta]$ with $m = \ln \sigma_1 + \ln \sigma_2$, $k = \ln \sigma_1 - \ln \sigma_2$ — standard, no reference needed
- **Currents:** E-071 §4.1's table (Somali 3.6 m/s, Gulf Stream ~ 2.5 , DWBC 0.02–0.1 m/s); RAPID 26.5°N array for the limb directions